

AIGR Gene Hypothesis Deep Research — Human A3GALT2 (UniProt U3KPV4)

Focus: function_assignment · **Hypothesis slug:** human-enzyme-constructs-and-glycan-substrate-specificity **Gene:** A3GALT2 (isoglobotriaosylceramide synthase / iGb3 synthase, iGb3S) · **Organism:** *Homo sapiens* (NCBITaxon:9606)

Executive Judgment

Verdict: Partially supported — but the enzymatic sub-claims are over-annotated.

The seed hypothesis bundles three separable claims: (1) human A3GALT2 retains glycosyltransferase/hexosyltransferase activity, (2) it contributes to glycosphingolipid biosynthesis, and (3) it has LacNAc-specific (N-acetyllactosaminide 3- α -galactosyltransferase, GO:0047276) activity. When each is evaluated **separately**, as the seed itself demands, the evidence does not support human catalytic activity. Every positive activity annotation on U3KPV4 rests on inference from active rodent orthologs (ECO:0000250 "by similarity", ISS, IEA, IBA), while the **only direct human experimental evidence is negative**: a chimera bearing the human A3GALT2 catalytic domain cannot synthesize iGb3, and the human-type Asn253 residue eliminates activity when transplanted into the otherwise-active rat enzyme ([PMID: 18630988](#)). This IDA-based **NOT|enables GO:0001962** (α -1,3-galactosyltransferase activity) is therefore well justified and should be **retained**, not erased.

The seed correctly instructs the curator to "interpret the existing NOT rather than erase it," and to treat "absence of an endogenous human assay" as distinct from "proof of inactivity." That epistemic caution is appropriate in principle — but here we have more than absence-of-assay: we have a **direct positive experiment showing loss of activity** (the Y253N substitution ablates the rat enzyme) plus corroborating comparative genomics showing that all human GT6-family genes except ABO are nonfunctional ([PMID: 17298992](#)). The "reconstructed-human-ORF weak positive in CHO cells" that the seed attributes to [PMID: 23378701](#) **could not be confirmed**: that paper's abstract concerns pig organs and reports the *absence* of iGb3/isoglobotriaosylceramide series glycosphingolipids by sensitive mass spectrometry; it provides no human endogenous

positive assay. The seed's framing of a "human-domain chimera negative vs. reconstructed-human-ORF weak positive" conflict thus appears to rest on a mis-attributed or unverifiable positive.

Most important caveats: (a) Human A3GALT2 is transcribed (GTEx v8, max 1.01 TPM in testis) and encodes an intact, foldable GT6 protein retaining DXD/metal-binding residues — so it is not a classical pseudogene; "over-annotated activity" is the correct framing, not "gene does not exist." (b) The LacNAc claim (GO:0047276) is a *different substrate* from the iGb3-synthase reaction assayed in [PMID: 18630988](#); no human assay directly tested LacNAc, so its removal/NOT-qualification is an inference-consistency lead rather than a directly assayed negative.

Key Findings

Finding 1 — Human A3GALT2 carries an inactivating Asn253 substitution; NOT GO:0001962 is directly supported

The reviewed UniProt entry U3KPV4 (340 aa) carries a CAUTION comment citing [PMID: 18630988](#). In that study, a chimeric molecule bearing the **human A3GALT2 catalytic domain was unable to synthesize iGb3**, and introducing the human-type residue (Tyr→Asn) at position 253 into the otherwise-active rat enzyme **completely eliminated α -1,3-galactosyltransferase activity**. The verified abstract quotes are direct:

"These hybrid molecules were unable to synthesize iGb3, due to at least one amino acid substitution." "our data suggest that iGb3S is not expressed in humans, and even if it were expressed, this enzyme would be inactive."

To confirm the residue identity independently, a Needleman–Wunsch global alignment was run between human U3KPV4 and the functional rat ortholog Q3V1N9 (69.4% identity over 340 aligned columns). The alignment confirms that **human position 253 = Asn (N)** where the active rat enzyme has **Tyr (Y)** — exactly the inactivating substitution described experimentally. QuickGO records the corresponding human annotation as **NOT|enables GO:0001962** (α -1,3-galactosyltransferase activity), evidence code **IDA**, sourced from

P 18630988

and assigned by UniProt. This is the single strongest, most direct piece of evidence in the whole dossier, and it is negative for catalytic activity. The NOT annotation is a faithful, experimentally grounded curation decision.

Finding 2 — All positive activity annotations (including LacNAc GO:0047276) are inference-only and conflict with the direct-assay NOT

A systematic audit of the evidence codes behind U3KPV4's positive activity statements shows a uniform pattern of **inference from rodent orthologs, never human experiment**:

Annotation	GO / RHEA	Evidence code	Source
iGb3 synthase FUNCTION statement	—	ECO:0000305 (curator inference)	 23378701
Catalytic activity (iGb3 reaction)	—	ECO:0000250 (by similarity)	rodent ortholog
Catalytic activity — LacNAc reaction	RHEA:13013 (LacNAc + UDP-Gal → Gal- α -1,3-LacNAc)	ECO:0000250 (by similarity)	rodent ortholog
enables GO:0047276 (N-acetyllactosaminide 3- α -galactosyltransferase)	—	ISS (GO_REF:0000024) + IEA (GO_REF:0000120)	electronic
enables GO:0016757 (glycosyltransferase)	—	IBA	phylogenetic
enables GO:0016758 (hexosyltransferase)	—	IEA	electronic
involved_in GO:0006688 (glycosphingolipid biosynthetic process)	—	IBA	phylogenetic
NOT	enables GO:0001962 (α -1,3-GalT activity)	—	IDA

The asymmetry is decisive: the *only* experimental (IDA) annotation is the negative one, while every positive annotation is "by similarity"/ISS/IEA/IBA propagated from active rodent iGb3 synthases. Critically, the FUNCTION statement cites [PMID: 23378701](#), whose verified abstract actually reports the **absence** of the product:

"We did not detect iGb3 or other isoglobo-series glycosphingoli[pids]"

That paper examined **pig organs used for xenotransplantation** and provides no human endogenous positive assay. Therefore the seed's "LacNAc-specific activity" claim (GO:0047276) is unsupported by any human data and is in direct logical tension with the NOT GO:0001962 — the LacNAc reaction is itself an α -1,3-galactosyltransferase reaction on a specific acceptor, and the enzyme demonstrated to have lost α -1,3-Gal transfer cannot be assumed to retain it selectively on LacNAc.

Finding 3 — Comparative genomics: human GT6-family genes are nonfunctional except ABO

The CAZy GT6 (ABO) glycosyltransferase family — comprising the A/B histo-blood-group enzymes (ABO), the Forssman synthase (GBGT1), the α Gal-epitope synthase (GGTA1), and the iGb3 synthase (A3GALT2/iGb3S) — was analyzed across fishes-to-mammals by Turcot-Dubois et al. ([PMID: 17298992](#)). The verified conclusions:

"In human, except ABO functional alleles, all other GT6 genes are either absent or nonfunctional." "Functional glycosyltransferase 6 (GT6) family members catalyze the transfer of galactose or N-acetylgalactosamine in alpha1,3 linkage to various substrates and synthesize structures related to the A and B histo-blood group antigens, the Forssman antigen, alphaGal epitope, and iGb3 glycolipid."

Human GGTA1 is a well-known pseudogene, and human GBGT1 is largely inactive; A3GALT2 falls in the same nonfunctional set. This evolutionary context both (a) reinforces the loss-of-function call and (b) flags the paralog-confusion risk behind the LacNAc annotation: GGTA1 makes the α Gal epitope on LacNAc acceptors, so an ISS/IEA propagation onto A3GALT2 for GO:0047276 may reflect family-level acceptor inference rather than a demonstrated human A3GALT2 activity. The presence of retained DXD/metal-binding residues (positions 199, 201)

alongside the inactivating acceptor/catalytic substitution (Asn253) is fully consistent with a **folded GT6 protein that has lost its specific α -1,3-Gal transferase activity** rather than a degraded pseudogene product.

Finding 4 — GTEx shows low but real transcription; it does not rescue the activity hypothesis

Because the 2008 study reported that spliced iGb3S mRNA was "not detected" in human tissues, modern RNA-seq was checked to update that claim. GTEx v8 median expression for A3GALT2 (ENSG00000184389.9) across 54 tissues shows **very low but non-zero transcription**: maximum median = **1.01 TPM (testis)**, next highest Brain_Cerebellum 0.43 and Whole_Blood 0.40; the across-tissue median is 0.12 TPM, and only 1 of 54 tissues exceeds 1 TPM. This partially updates the 2008 "not detected" statement — the locus is transcribed at trace levels — but does not rescue the enzymatic hypothesis, because (a) expression is uniformly very low and (b) the encoded protein still carries the experimentally validated inactivating Asn253 residue. Transcription of an mRNA encoding a catalytically dead protein is not evidence of enzymatic function.

Mechanistic Model / Interpretation

The molecular question is narrow and well posed: does the human A3GALT2 protein directly catalyze transfer of galactose in α -1,3 linkage (to Gal- β -1-4-GlcNAc/LacNAc, or to lactosylceramide en route to iGb3)? The evidence converges on **"no."**

ACTIVE ORTHOLOG (rat Q3V1N9)

GT6 fold + DXD/metal sitesame....

Acceptor/catalytic residue Tyr253

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LacCer + UDP-Gal --> iGb3 (works)

LacNAc + UDP-Gal --> Gal- α 1,3-LacNAc

HUMAN A3GALT2 (U3KPV4)

GT6 fold + DXD/metal site (res 199,201 retained)

Acceptor/catalytic residue Asn253 <-- INACTIVATED

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v

LacCer + UDP-Gal --> iGb3 (FAILS, https://pubmed.ncbi.nlm.nih.gov/16111111/

LacNAc reaction: never assayed in human;
inferred "by similarity" only (RHEA:13013)

Direct human evidence: NOT|enables GO:0001962 (IDA, https://pubmed.ncbi.nlm.nih.gov/16111111/

All positive terms: ECO:0000250 / ISS / IEA / IBA

<== inference from rodent

The interpretive key is the distinction the seed itself insists on: **a positive on a different substrate does not establish LacNAc activity, and absence of an endogenous assay is not proof of inactivity**. Applying that rigor symmetrically:

- The **negative** here is *not* mere absence of an endogenous assay — it is a *positive experiment demonstrating loss*: transplanting the human Asn253 into the active rat enzyme abolishes activity (a gain-of-loss experiment), and the human catalytic domain chimera fails to make iGb3.
- The **LacNAc positive** is exactly the kind of "different-substrate / by-similarity" claim the seed warns against — it is propagated from rodent orthologs and from family paralogs (GGTA1), never measured on human A3GALT2.

Hence the coherent model is: human A3GALT2 is a **transcribed, foldable, but catalytically dead** relic of an ancestral GT6 iGb3 synthase, inactivated by the Tyr→Asn substitution at the acceptor/catalytic position 253, consistent with the family-wide loss of GT6 function in humans (except ABO). The "iGb3 synthase" and "LacNAc 3- α -galactosyltransferase" molecular-function annotations describe the **ancestral/ortholog activity**, not the extant human protein.

Evidence Matrix

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Co
PMID: 18630988	Direct assay (chimera + point mutant)	Refutes activity	Does human A3GALT2 catalytic domain make iGb3 / retain α -1,3-GalT?	Human- domain chimera cannot synthesize iGb3; human- type Y253N ablates rat enzyme activity	Human catalytic domain; rat enzyme scaffold; in vitro	Hi Ch co fu en hu en na re po ga
U3KPV4 / QuickGO (from  18630988)	Database (IDA- backed NOT)	Supports NOT	Is the NOT GO:0001962 experimentally grounded?	NOT	enables GO:0001962, evidence IDA	Hi Un as
Sequence alignment (U3KPV4 vs rat Q3V1N9)	Structural/ evolutionary (computational)	Supports loss	Does human carry the inactivating residue?	69.4% identity; human pos 253 = Asn vs rat Tyr	In silico Needleman- Wunsch	Hi re id fu ef in fr 
PMID: 17298992	Structural/ evolutionary (comparative genomics)	Supports loss / qualifies LacNAc	Are human GT6 genes functional?	All human GT6 genes nonfunctional except ABO; family transfers Gal/ GalNAc in α 1,3	Fishes→mammals; human genome	Hi fa co in A3 sp fla pa co La

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Co
PMID: 23378701	Direct assay (MS)	Competing / qualifies	Does the seed's "reconstructed-human-ORF weak positive" exist?	Reports <i>absence</i> of iGb3/isogloboseries GSLs by sensitive MS; no human endogenous positive assay	Pig organs (xenotransplant)	Un se cla po ab pi ci Un FU no
PMID: 17372206	Mutant phenotype (KO mouse)	Qualifies (BP relevance)	Is iGb3 a required physiological ligand?	iGb3S-KO mice develop normally; iGb3 unlikely to be endogenous NKT-selecting ligand	Mouse	Co co do bi hu ca ac
GTEx v8	Database (expression)	Qualifies	Is A3GALT2 transcribed in human?	Max median 1.01 TPM (testis); median 0.12 TPM; 1/54 tissues >1 TPM	Human, 54 tissues	Hi tra irn ca co

GO Curation Implications

Lead requiring curator verification. The evidence favors the following actions:

1. **Retain NOT | enables GO:0001962** (α -1,3-galactosyltransferase activity). This is the correctly grounded, IDA-backed decision from [PMID: 18630988](#), corroborated by the Asn253 residue and family-wide GT6 loss. Do **not** erase it. (*MF term; retain as NOT.*)

2. **Re-scope the positive MF activity terms as ortholog-inference, not extant human function.** GO:0047276 (N-acetyllactosaminide 3- α -galactosyltransferase, the LacNAc term), the iGb3-synthase catalytic activity, GO:0016757 (glycosyltransferase, IBA), and GO:0016758 (hexosyltransferase, IEA) are all inference-only. Because the LacNAc reaction is itself an α -1,3-Gal transfer, GO:0047276 is in logical tension with the NOT and is a strong candidate for **removal or a NOT qualifier**, pending curator confirmation that no human LacNAc assay exists. (*MF; generalize/remove or NOT-qualify.*)
3. **Treat BP GO:0006688 (glycosphingolipid biosynthetic process, IBA) as non-core / inference-only** for the human gene. The human protein does not demonstrably contribute product to this pathway. (*BP; downgrade to non-core or remove.*)
4. **Do not recommend "protein binding" as a fallback** — the informative statement is that the human ortholog *lost* a specific catalytic MF, which is better captured by the retained NOT plus a CAUTION note than by a vague binding term.

GO decision table

GO term	Current	Evidence	Recommended lead
GO:0001962 α -1,3-galactosyltransferase	NOT	enables (IDA)	Direct human negative (P 18630988)
GO:0047276 LacNAc 3- α -GalT	enables (ISS/IEA)	Inference only; conflicts with NOT	Remove or NOT-qualify
iGb3 synthase catalytic activity	enables (ECO:0000250)	By similarity	Re-scope as ortholog inference
GO:0016757 glycosyltransferase	enables (IBA)	Phylogenetic	Generalize/annotate as inferred; low priority
GO:0016758 hexosyltransferase	enables (IEA)	Electronic	Non-core
GO:0006688 glycosphingolipid biosynthesis	involved_in (IBA)	Phylogenetic	Downgrade to non-core / inference

Mechanistic Scope

The immediate molecular function under test is **UDP-galactose:acceptor α -1,3-galactosyltransferase activity** — specifically iGb3 synthesis (Gal- α -1,3 onto lactosylceramide) and, per the seed, the LacNAc-acceptor variant (RHEA:13013). This is a single enzymatic step.

The direct evidence

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P 18630988

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speaks to exactly this step and finds it lost in the human sequence.

Downstream/contextual layers that must **not** be conflated with the molecular activity: - **iGb3 as a glycosphingolipid** and its proposed role as an endogenous NKT-cell-selecting lipid — [PMID: 17372206](#) shows iGb3S-KO mice develop normally with normal iNKT cells, arguing iGb3 is not the required ligand. This is a BP/physiology question, not evidence for human catalytic activity. - **Xenotransplantation glycobiology** ([PMID: 23378701](#)) concerns product abundance in pig organs, again downstream of the human molecular-function question. - **Transcription** (GTEx) is upstream of protein function and does not imply catalysis.

Keeping these separate is essential: the human gene may be transcribed and its ortholog physiologically studied, yet the *human protein's direct catalytic MF* is the blinded decision, and it is negative.

Conflicts and Alternatives

- **Ortholog-inference over-annotation (primary alternative, favored).** The positive terms are propagated from active rodent iGb3 synthases; they describe ancestral function, not human function. This is the parsimonious reading of the ECO:0000250/ISS/IEA/IBA pattern.
- **Paralog confusion for LacNAc (GO:0047276).** Within GT6, GGTA1 synthesizes the α Gal epitope on LacNAc. Family-level acceptor inference could have seeded the LacNAc annotation onto A3GALT2 without direct measurement — precisely the "different substrate does not establish LacNAc activity" caution in the seed.
- **Database carry-over.** The FUNCTION statement cites

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(a pig-organ negative), suggesting the positive framing is a citation/carry-over artifact rather than a demonstrated human positive.

- **The seed's asserted "reconstructed-human-ORF weak positive in CHO cells."** This could not be verified against the cited abstract (which is a pig-organ negative). If such a CHO reconstitution positive exists, it would be a genuine competing datum and must be located in the full text; on current evidence it appears mis-attributed. A weak in-CHO signal, if real, would still be an in-vitro overexpression result requiring reconciliation with the human-domain chimera negative and the Asn253 point-mutant result.
- **Chimera caveat (pro-hypothesis nuance).**

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tested a human *catalytic domain* in a chimeric/rat scaffold, not the intact native human enzyme in human cells. This is the strongest pro-hypothesis point — but the accompanying single-residue Y253N gain-of-loss experiment substantially closes that gap.

Limitations and Knowledge Gaps

1. **No intact-native-human-enzyme assay.** The direct negative uses a chimera plus a point mutant, not full-length human A3GALT2 expressed and assayed in human cells against a panel of acceptors. *Why it matters:* the seed's epistemics ("absence of endogenous assay $\neq$ inactivity") apply. *Resolver:* express full-length U3KPV4 in a null background and assay UDP-Gal transfer to LacCer and LacNAc.
2. **LacNAc acceptor never directly tested in human.** GO:0047276 rests entirely on inference. *Resolver:* in-vitro glycosyltransferase assay with LacNAc + UDP-Gal (RHEA:13013) using purified human protein; LC-MS product detection.
3. **The seed's CHO "weak positive" is unverified.** *Why it matters:* it is the linchpin of the claimed conflict. *Resolver:* obtain

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full text (and any companion) to confirm/deny a human-ORF reconstitution result; search for the primary source of any CHO positive.

4. **Functional effect of Asn253 confirmed only via the rat scaffold.** *Resolver:* reverse experiment — Asn253→Tyr "resurrection" of human A3GALT2 to test whether a single back-mutation restores activity, which would cleanly prove the residue is the sole inactivating lesion.

5. **Trace testis transcription of unknown consequence.** *Resolver:* ribosome-profiling/ proteomics evidence for translated, stable human A3GALT2 protein.

Discriminating Tests

1. **Full-length human A3GALT2 in-vitro assay** against LacCer *and* LacNAc with UDP-Gal; LC-MS/MS product quantification. Distinguishes "catalytically dead" from "active on some acceptor" and directly tests GO:0047276.
2. **Asn253→Tyr resurrection mutant** of human A3GALT2: if activity is restored, it nails the single-residue inactivation model and confirms the ancestral specificity.
3. **Reciprocal residue swaps across GT6 paralogs** (A3GALT2 ↔ GGTA1 acceptor loops) to test the paralog-confusion hypothesis for LacNAc specificity.
4. **Targeted proteomics / ribo-seq** in testis to establish whether the trace transcript yields stable protein.
5. **Literature/full-text retrieval** of

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and citing papers to locate or refute the seed's "reconstructed-human-ORF weak positive in CHO cells."

Proposed Follow-up Experiments / Actions (Curation Leads)

Leads requiring curator verification:

- **Action lead A (retain):** Keep NOT | enables GO:0001962 (IDA,

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Snippet to verify: "these hybrid molecules were unable to synthesize iGb3, due to at least one amino acid substitution" and "even if it were expressed, this enzyme would be inactive."

- **Action lead B (remove / NOT-qualify):** Flag GO:0047276 (LacNAc 3- α -GalT) as inference-only and in tension with the NOT; recommend removal or NOT-qualification pending a

human LacNAc assay. *Rationale*: ISS/IEA only; the LacNAc reaction is an α -1,3-Gal transfer, which the human enzyme has lost.

- **Action lead C (downgrade)**: Treat **GO:0006688** (glycosphingolipid biosynthetic process, IBA) and **GO:0016757/GO:0016758** as non-core / ortholog-inference for the human gene.
- **Reference-integrity lead D**: Note that the FUNCTION statement cites

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whose abstract reports *absence* of isoglobo-series products in pig organs and provides **no human positive assay**. *Snippet to verify*: "We did not detect iGb3 or other isoglobo-series glycosphingoli[pids]." Curator should confirm whether this citation actually supports a human FUNCTION.

- **Reconciliation lead E**: Attempt to locate the seed's "reconstructed-human-ORF weak positive in CHO cells." If unlocatable, record that the claimed

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conflict does not resolve into a genuine human positive.

- **Suggested question for curators**: Should A3GALT2's human MF annotations be reframed uniformly as "ancestral/ortholog activity, lost in *H. sapiens*," with a single CAUTION note, rather than carrying multiple positive activity terms alongside the NOT?
- **Suggested experiments**: full-length human assay (LacCer + LacNAc), Asn253→Tyr resurrection, testis proteomics — as detailed in Discriminating Tests.

Evidence Base (Literature)

- **PMID: 18630988** — *Humans lack iGb3 due to the absence of functional iGb3-synthase: implications for NKT cell development and transplantation*. The pivotal direct-assay negative: human catalytic-domain chimera cannot make iGb3; human-type Y253N ablates the active rat enzyme. Grounds the NOT GO:0001962.
- **PMID: 23378701** — *Lack of iGb3 and Isoglobo-Series Glycosphingolipids in Pig Organs Used for Xenotransplantation: Implications for Natural Killer T-Cell Biology*. Reports absence of isoglobo-series products by sensitive MS in pig organs; cited by UniProt for FUNCTION but provides no human positive assay — undercuts the seed's claimed positive.

- **PMID: 17298992** — *Long-term evolution of the CAZY glycosyltransferase 6 (ABO) gene family from fishes to mammals — a birth-and-death evolution model.* Establishes that all human GT6 genes except ABO are nonfunctional and defines the paralog set (GGTA1/αGal-on-LacNAc, GBGT1/Forssman, A3GALT2/iGb3), relevant to paralog-confusion in the LacNAc annotation.
- **PMID: 17372206** — *Normal development and function of invariant natural killer T cells in mice with isoglobotrihexosylceramide (iGb3) deficiency.* iGb3S-KO mice are phenotypically normal; iGb3 unlikely to be the endogenous NKT-selecting ligand — contextualizes the BP/physiology layer, separate from human catalytic MF.

Prepared for AI Gene Review curation. All activity conclusions distinguish direct human experimental evidence (negative, IDA) from ortholog/paralog inference (positive, ECO:0000250/ISS/IEA/IBA). Computational residue identity (Asn253) was verified by pairwise alignment; the sequence conflict, expression, and evidence-code audits are reported conservatively as leads for curator verification.

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